Darshan UNIVERSITY

योग: कर्मसु कौशलम्

Python Programming - 2301CS404

Lab - 14

Roll No.:324

Name:Aditya Chauhan

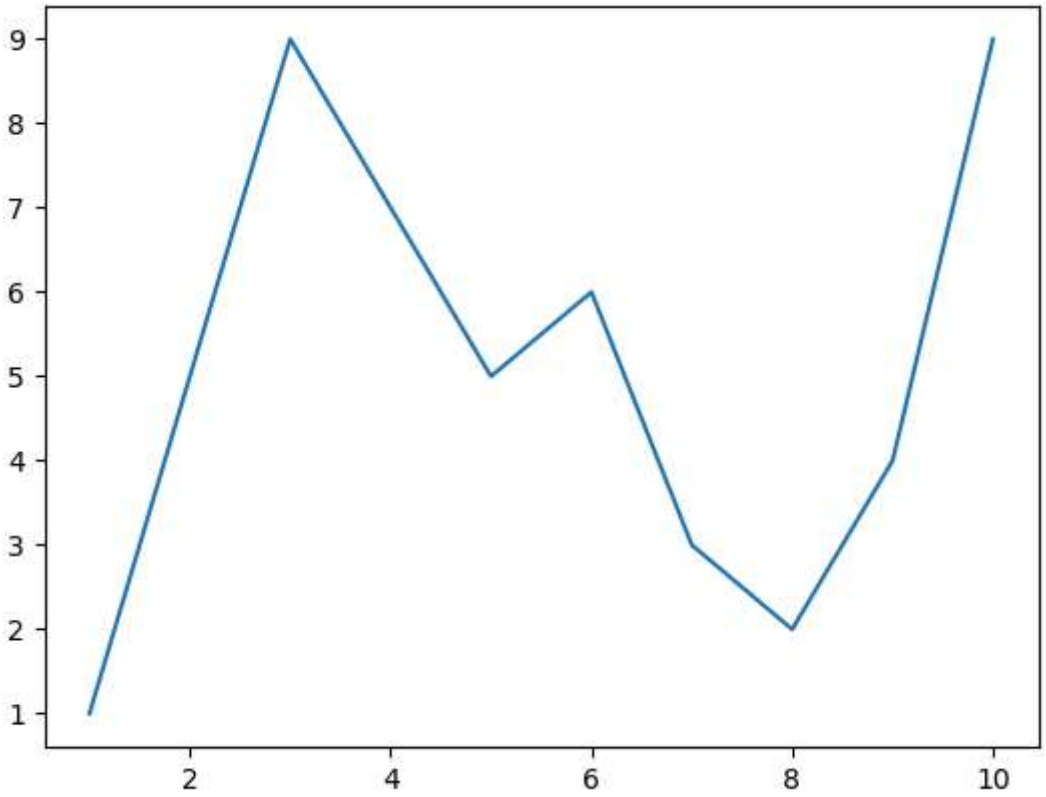
Import matplotlib below

In [1]:
```
import matplotlib.pyplot as plt
```

01) WAP to display the line chart for given data for x and y axis.

In [2]:
```
x = range(1,11)
y = [1,5,9,7,5,6,3,2,4,9]
```

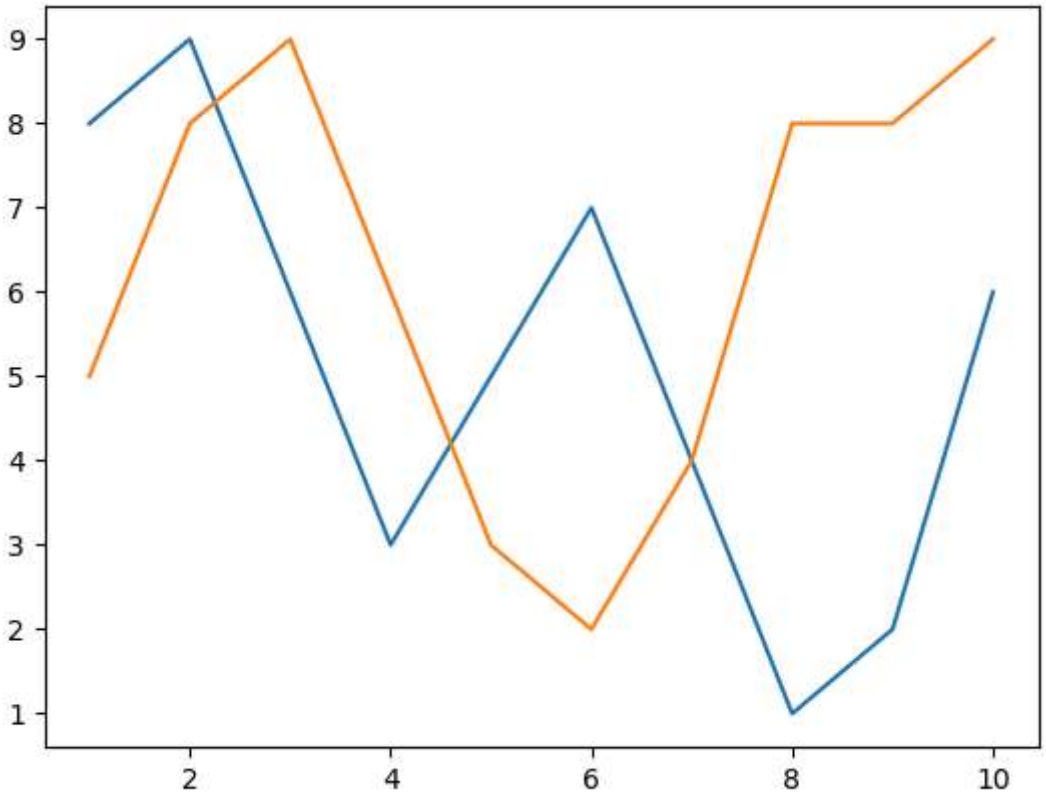
In [3]:
```
plt.plot(x,y)
plt.show()
```



02) WAP to generate line chart for given data.

In [4]:
```
x = range(1,11,1)
cxMarks= [8,9,6,3,5,7,4,1,2,6]
cyMarks= [5,8,9,6,3,2,4,8,8,9]
```

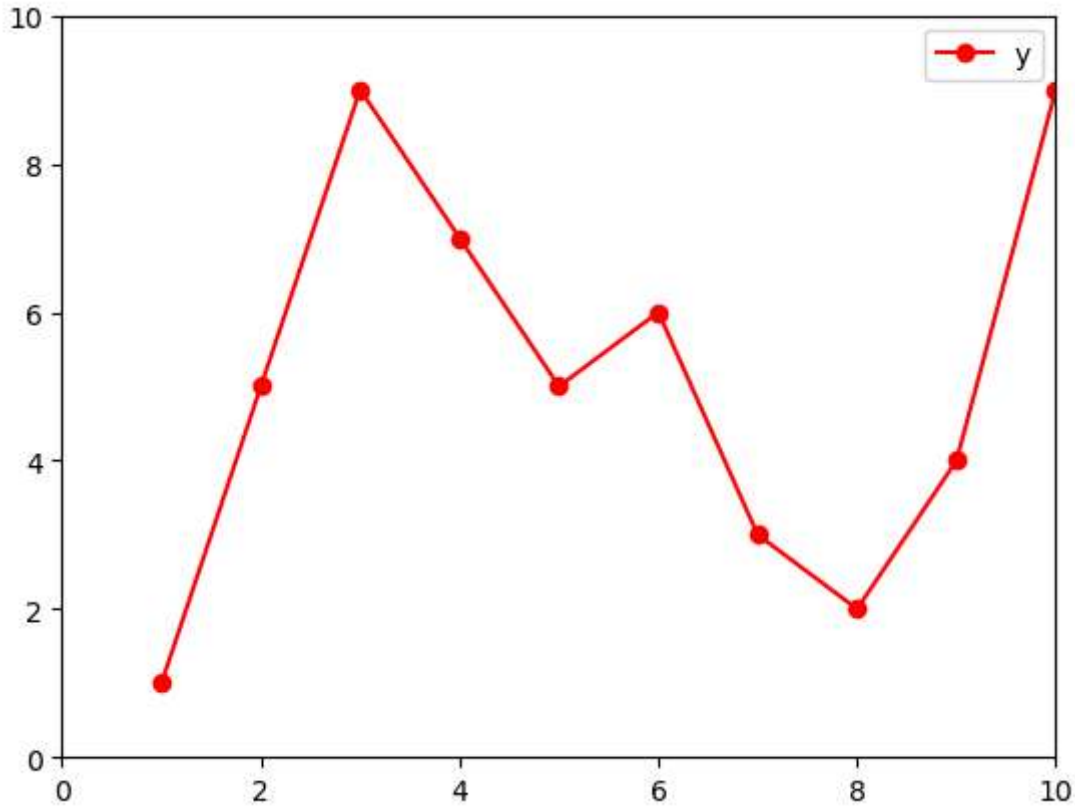
In [5]:
```
plt.plot(x,cxMarks)
plt.plot(x,cyMarks)
plt.show()
```



03) WAP to generate line chart for given data, but create your custom axis.

In [1]:
```
x = range(1,11)
y = [1,5,9,7,5,6,3,2,4,9]
```

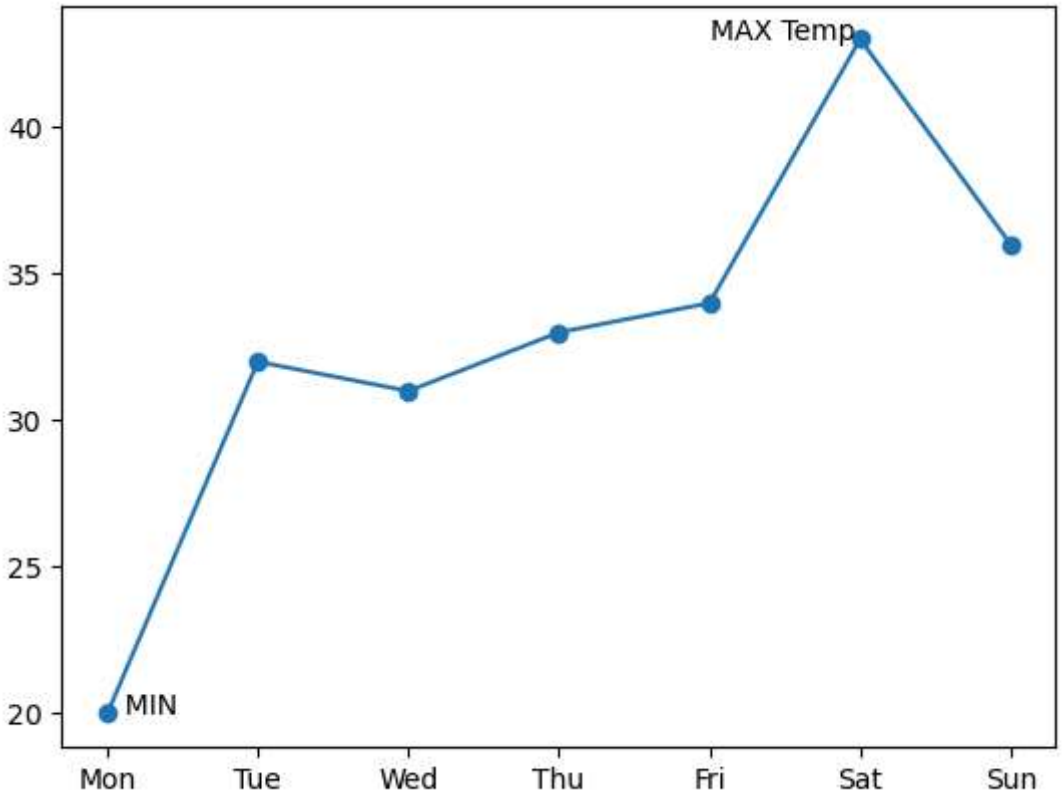
In [10]:
```
plt.plot(x,y,c="r",label="y",marker="o")
plt.xlim(0,10)
plt.ylim(0,10)
plt.legend()
plt.show()
```



04) WAP to plot a line chart showing temperature variation during a week. Mark the lowest and highest temperatue.

In [8]:
```
Days = ["Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"]
Temperature = [20, 32, 31, 33, 34, 43, 36]
```

In [25]:
```
plt.plot(Days,Temperature,marker="o")
plt.annotate("MAX Temp",xy=["Fri",max(Temperature)])
plt.annotate("  MIN",xy=["Mon",min(Temperature)])
plt.show()
```

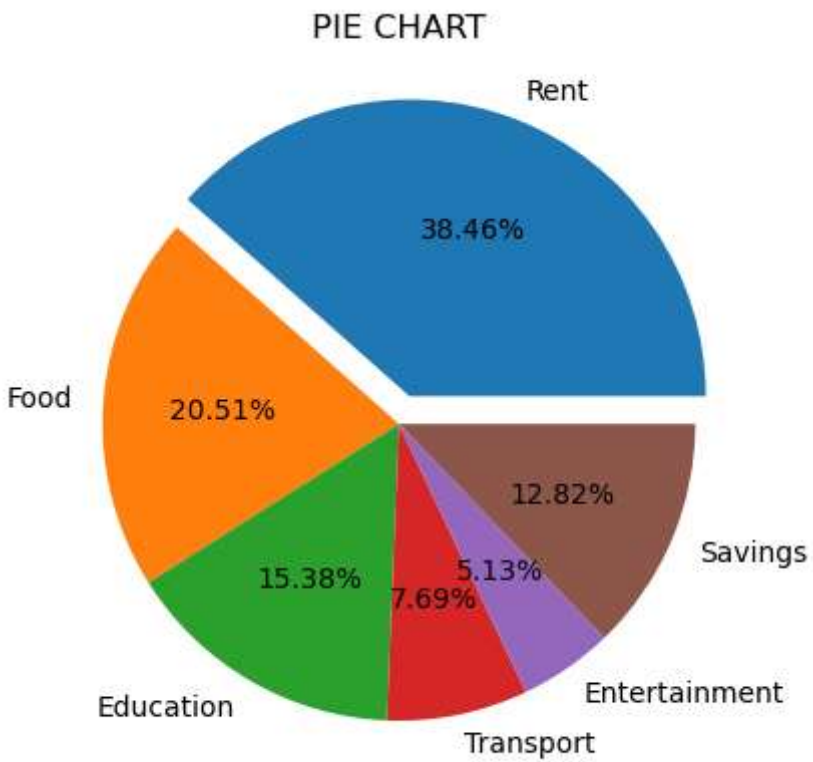


05) WAP to create a detailed pie chart representing the monthly expenditure distribution of a family.

The chart should display labels, percentage values, title, different colors for each slice, and explode one slice to highlight the highest expense.

In [26]:
```
Categories = ["Rent", "Food", "Education", "Transport", "Entertainment", "Savings"]
Amount = [15000, 8000, 6000, 3000, 2000, 5000]
```

In [38]:
```
plt.pie(Amount,labels = Categories,autopct="%.2f%%",explode=[0.1,0,0,0,0,0])
plt.title("PIE CHART")
plt.show()
```

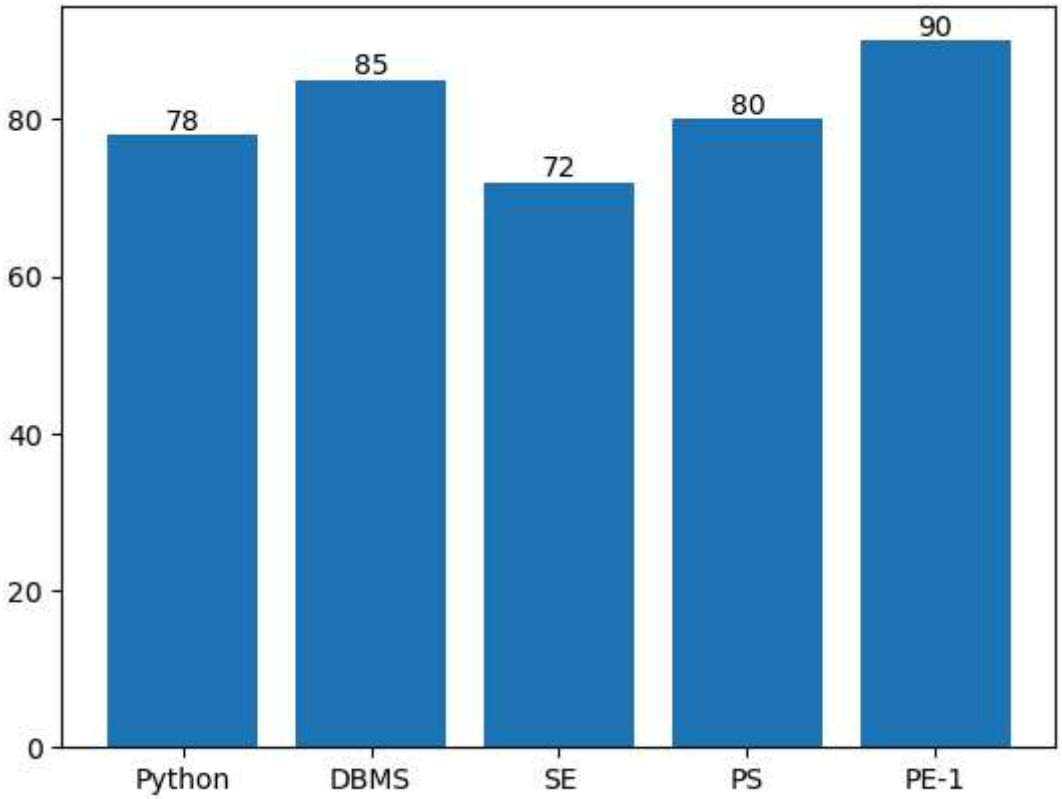


06) WAP to create bar chart showing the marks obtained by students in different subjects.

The program should also display the marks on the top of the respective bar.

In [39]:
```
Subjects = ["Python", "DBMS", "SE", "PS", "PE-1"]
Marks = [78, 85, 72, 80, 90]
```

In [48]:
```
bar = plt.bar(Subjects,Marks)
plt.bar_label(bar,labels = Marks)
plt.show()
```

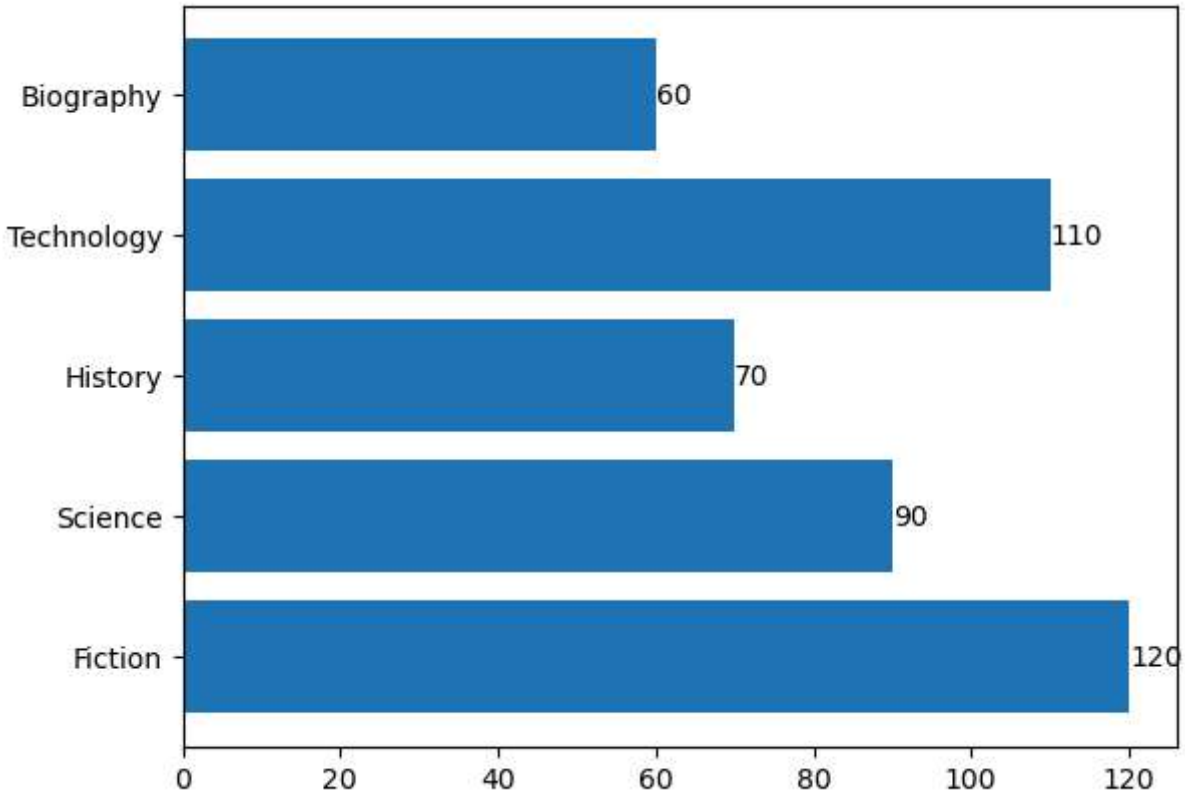


07) WAP to create a horizontal bar chart representing the number of books sold in different categories.

The program should also display the number of books sold at the end of the respective bar.

In [3]:
```
Categories = ["Fiction", "Science", "History", "Technology", "Biography"]
Books_Sold = [120, 90, 70, 110, 60]
```

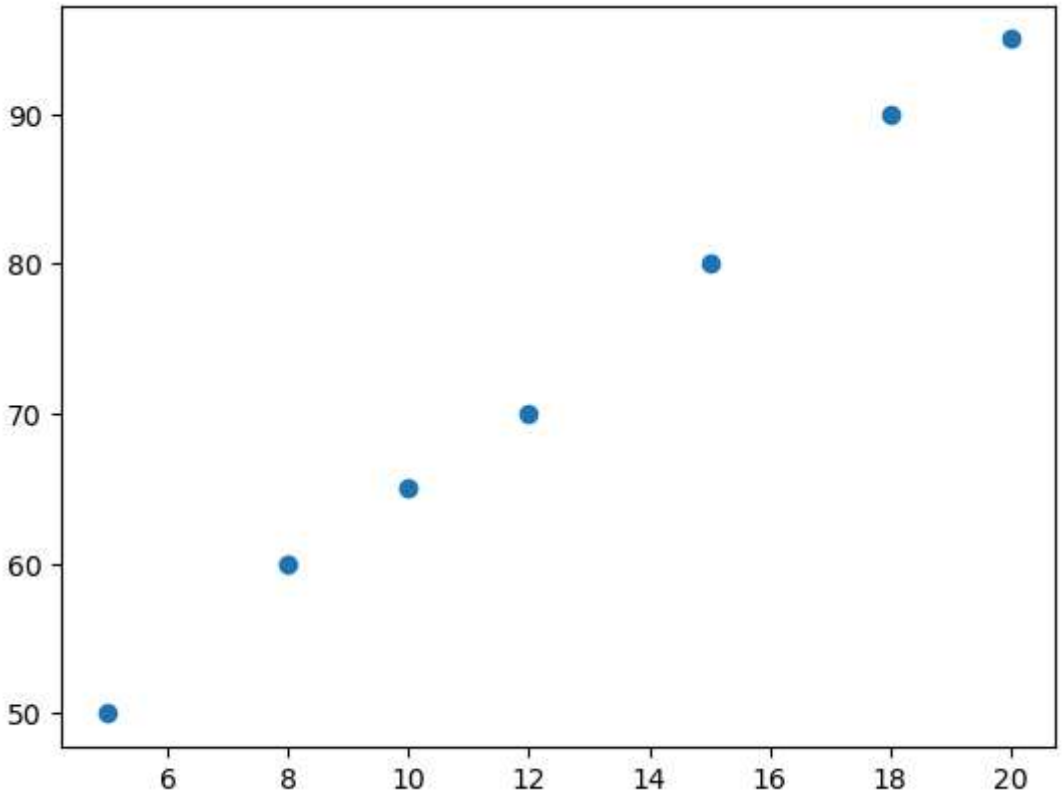
In [5]:
```
bars = plt.barh(Categories,Books_Sold)
plt.bar_label(bars,labels=Books_Sold)
plt.show()
```



08) WAP to create a scatter plot representing the relationship between advertising expenditure and sales revenue of a company.

In [7]:
```
Advertising_Cost = [5, 8, 10, 12, 15, 18, 20]
Sales_Revenue = [50, 60, 65, 70, 80, 90, 95]
```

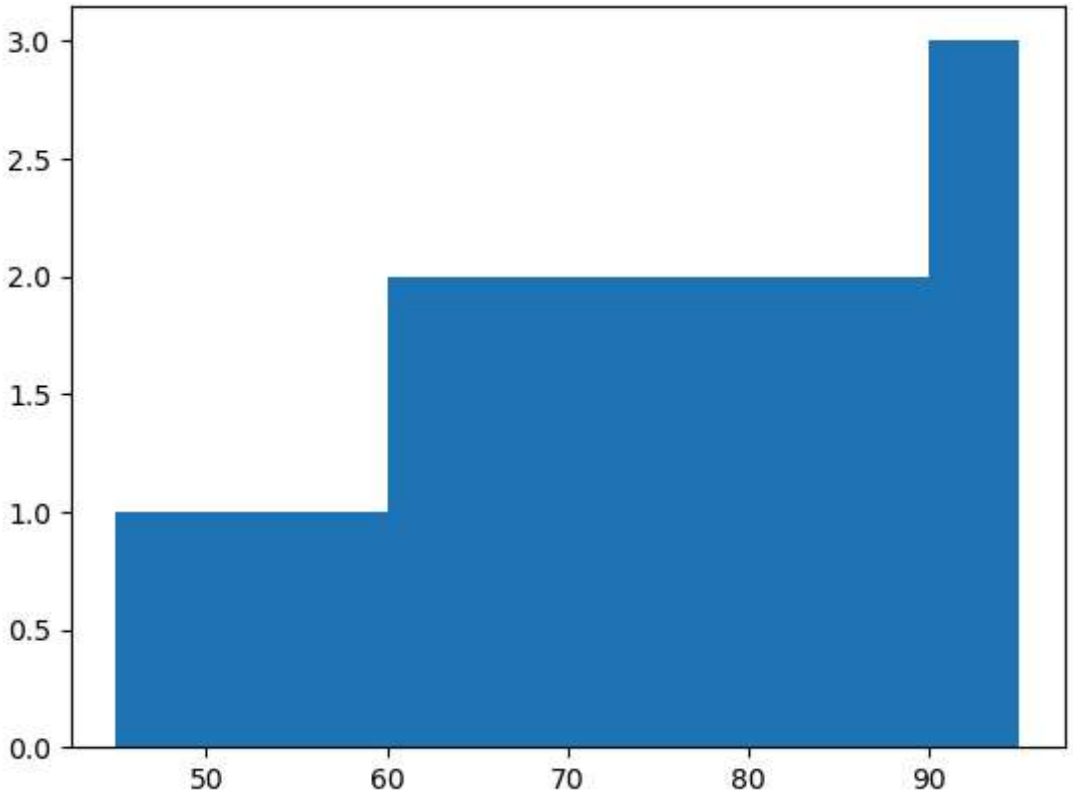
In [8]:
```
plt.scatter(Advertising_Cost,Sales_Revenue)
plt.show()
```



09) WAP to create a histogram representing the distribution of marks obtained by students in a class.

In [9]:
```
Marks = [45, 50, 55, 60, 62, 65, 68, 70, 72, 75, 78, 80, 82, 85, 88, 90, 92, 95]
```

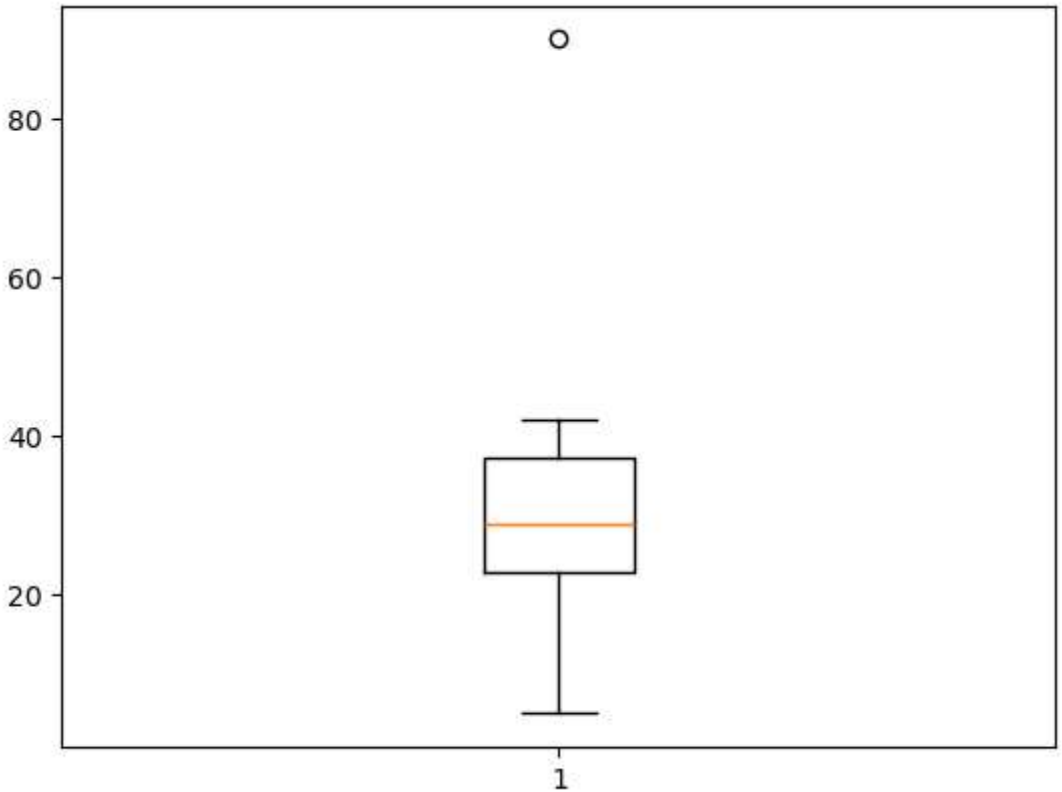
In [10]:
```
plt.hist(Marks)
plt.show()
```



10) WAP to create a box plot representing the distribution of daily commuting time (in minutes) of employees and identify outliers (if any).

In [11]:
```
Commuting_Time= [5, 18, 20, 22, 25, 27, 28, 30, 32, 35, 38, 40, 42, 90]
```

In [12]:
```
plt.boxplot(Commuting_Time)
plt.show()
```



In [ ]: